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Sebnem Sara Yayla, MSc

Data Scientist



Data Scientist with a strong foundation in machine learning, statistical modeling, and data-driven research. Experienced in building end-to-end analytical pipelines in Python and R, from preprocessing and feature engineering to model development and deployment. Skilled at turning complex datasets into actionable insights and motivated by solving real-world problems through reproducible analytics.

SKILLS

Programming & Scripting	Python, SQL, R
Data Science & ML	scikit-learn, TensorFlow, PyTorch
Business Intelligence	Tableau Desktop, Tableau Prep, KNIME
Data Engineering	dbt, Fivetran, Apache Airflow, S3, Data Warehouse (MSSQL, BigQuery)
Development Tools	Git, Jupyter, Anaconda
Languages	English (C1), German (B1), Turkish (native)

EXPERIENCE

Research Project Data Scientist Mar 2024 — Jun 2025
FH Wiener Neustadt - Josef Ressel Zentrum Tulln, Austria

- Collected, processed, and analyzed large datasets, applying preprocessing, outlier detection, and resampling techniques to ensure data quality.
- Developed end-to-end machine learning pipelines in **Python (scikit-learn)** and **R** to predict material composition with strong regression performance.
- Applied advanced ML methods (**Random Forest, XGBoost, SVR, CNN**) to high-dimensional datasets.
- Applied statistical methods and visualized results to support research and development decisions.

BI Analyst Oct 2021 — Oct 2023
Use Data To Lead Vienna, Austria

- Built and maintained data pipelines for extracting, cleansing, and transforming large operational datasets using **Tableau Prep**.
- Developed interactive data visualizations and dashboards in **Tableau** to monitor performance metrics and identify trends.
- Executed **ELT** initiatives utilizing **Fivetran, dbt and BigQuery** for efficient data integration and transformation.

Analytics Consultant Jan 2019 — Jan 2021
invent.ai Istanbul, Turkey

- Developed and maintained machine learning models in **Python** to optimize inventory performance and reduce excess stock.
- Used **R and SQL** to conduct customer behavior analysis based on sales and inventory data to inform strategic business decisions.
- Supported production analytics workflows using **Apache Airflow and S3**, and managed structured data in **MSSQL** data warehouses.

IT Auditor Oct 2017 — Dec 2018
KPMG Istanbul, Turkey

- Performed audits of information systems in financial institutions.
- Conducted journal voucher data analysis for suspicious data detection.

EDUCATION

M.Sc. in Data Science, *University of Vienna* Mar 2021 — Sep 2025

- Machine Learning, Mining Massive Data, Visual and Exploratory Data Analysis, Natural Language Processing

B.Sc. in Industrial Engineering, *Sabanci University* Oct 2012 — Jun 2017

- Stochastic Models, Supply Chain, Manufacturing Systems

PUBLICATION

Lilek, D.; Yayla, S.S.; et al. (2026).

NIR Spectroscopy and Machine Learning for the Quantification of Blended Textiles: Towards Improved Understanding for Textile Recycling

Applied Sciences (MDPI), 16(7), 3242.

CERTIFICATIONS

Developing Large Language Models

Feb 2026

Tableau Desktop Specialist Certification

Aug 2022

PROJECTS

Thesis Project - Supervised Machine Learning for the Quantification of Cotton/Polyester Textiles Using IR Spectroscopy

- Created **machine learning pipeline** using spectral data to predict cotton content in textile fiber, achieving an error of 1%.
- Demonstrated an efficient textile sorting approach, improving recycling efficiency.
- Built multiple regression and deep learning models, including **XGBoost, Random Forest, SVR, MLP, and CNN**.

Crowd-sourcing a text corpus for measuring linguistic prevalence

- Applied **web scraping** techniques to create a blog corpus.
- Processed and cleaned the text data with Python to enable the calculation of word prevalence.

Data Analysis of "Inside Airbnb" Listings in Vienna

- Developed predictive models with the KNIME platform to estimate property prices based on specific features.
- Achieved a commendable R-squared value of 57% through the utilization of a Gradient Boosting Machine Learner.

ACTIVITIES

Heritage Maintenance Camp Leader, *Offene Häuser E.V.*

Jul 2014 — Aug 2014

- Renovation of historical structures at different camps in Germany.

Volunteer Trainer, *AIESEC in Moscow*

Jul 2013 — Aug 2013

- Educational activities with children to fulfill their potential.

Team Leader, *AIESEC in Istanbul*

Jan 2013 — Jan 2014

- Serving as team leader to recruit students for international internships.

Team Member, *Sabanci University Civic Involvement Projects*

Oct 2012 — Jul 2013

- Provided weekly support to elderly residents in a care home contributing to their well-being.

REFERENCES

Christian Schimper

FH Wiener Neustadt - Josef Ressel Zentrum

Christian Artner-Yanni

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